Lab 8 Rijndael Algorithm

```
import java.util.*;
import java.security.*;
import javax.crypto.*;
import javax.crypto.spec.*;
import javax.xml.bind.DatatypeConverter;

public class lab8 {
    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);

        KeyGenerator kg = KeyGenerator.getInstance("AES");
        kg.init(128);
        SecretKey key = kg.generateKey();

        byte[] iv = new byte[16];
        new SecureRandom().nextBytes(iv);

         Cipher c = Cipher.getInstance("AES/CBC/PKCS5Padding");

        System.out.print("Enter message: ");
        String msg = sc.nextLine();

        c.init(Cipher.ENCRYPT_MODE, key, new IvParameterSpec(iv));
        byte[] encrypted = c.doFinal(msg.getBytes("UTF-8"));

        c.init(Cipher.DECRYPT_MODE, key, new IvParameterSpec(iv));
        byte[] decrypted = c.doFinal(encrypted);

        System.out.println("Key       : "
+DatatypeConverter.printBase64Binary(key.getEncoded()));

        System.out.println("IV        : "
+DatatypeConverter.printBase64Binary(iv));

        System.out.println("Encrypted : "
+DatatypeConverter.printBase64Binary(encrypted));

        System.out.println("Decrypted : " +new String(decrypted, "UTF-8"));

        sc.close();
    }
}
```